OOPS ASSIGNMENT 9

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1. Write a C++ program to write number 1 to 200 in a data file NUM.TXT.

Sol1:

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
  ofstream fout("NUM.TXT");

  for(int i = 1; i <= 200; i++) {
    fout << i << "\n";
  }

  fout.close();
  cout << "Numbers written successfully!";
  return 0;
}
```

Start here | ques2.cpp | ques1.cpp | NOTES.TXT | ques1actual.cpp | NUM.TXT

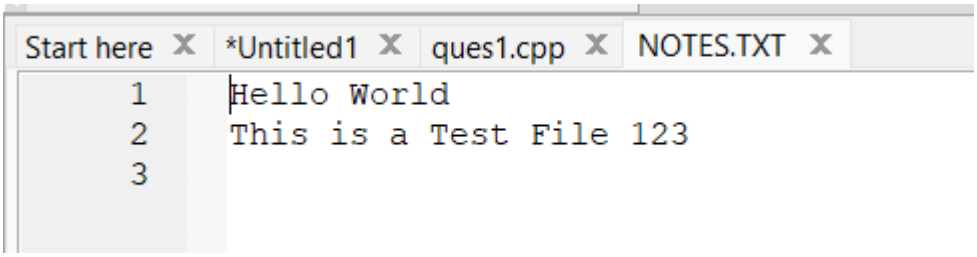
```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
```

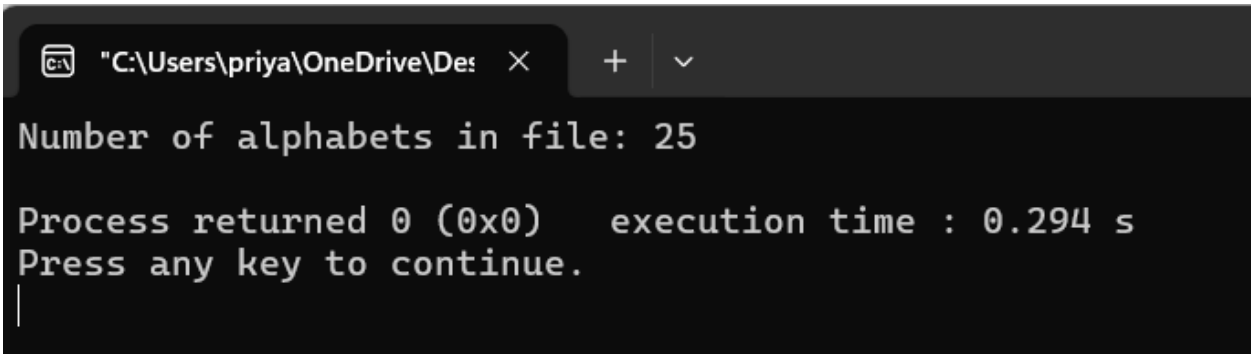
2. Write a user-defined function in C++ to read the content from a text file NOTES.TXT, count and display the number of alphabets present in it.

Sol2:

```
#include <iostream>

#include <fstream>

#include <cctype>   // for isalpha()

using namespace std;


// User-defined function

void countAlphabets() {

  ifstream file("NOTES.TXT");


  if (!file) {

    cout << "Error opening file!" << endl;

    return;

  }


  char ch;

  int count = 0;


  // Read file character by character

  while (file.get(ch)) {

    if (isalpha(ch)) {

      count++;

    }
```

```
    }

    file.close();

    cout << "Number of alphabets in file: " << count << endl;
}

int main() {
    countAlphabets();
    return 0;
}
```





3. Write a program to copy the contents of one file to another.

Sol 3:

```
#include <iostream>
```

```
#include <fstream>
using namespace std;

int main() {
  ifstream source("source.txt");  // file to read from
  ofstream destination("dest.txt"); // file to write into

  if (!source) {
    cout << "Error opening source file!" << endl;
    return 1;
  }

  if (!destination) {
    cout << "Error opening destination file!" << endl;
    return 1;
  }

  char ch;

  // Copy character by character
  while (source.get(ch)) {
    destination.put(ch);
  }

  cout << "File copied successfully!" << endl;
```

```
    source.close();
    destination.close();

    return 0;
}
```

Source file:

Start here | ques2.cpp | ques1.cpp | NOTES.TXT | ques1actual.cpp | NUM.TXT | ques3.cpp | ques5.cpp | source.txt | dest.txt

```
Hello World
This is a Test File 123
```

Destination file:

Start here | ques2.cpp | ques1.cpp | NOTES.TXT | ques1actual.cpp | NUM.TXT | ques3.cpp | ques5.cpp | source.txt | dest.txt

```
Hello World
This is a Test File 123
```

4. Write a program to implement I/O operations on characters. I/O operations include inputting a string, calculating the length of the string, storing the string in a file and fetching the stored characters from it.

Sol 4:

```
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

int main() {
    char str[100];

    cout << "Enter a string: ";
```

```
    cin.getline(str, 100);

    int len = strlen(str);
    cout << "Length = " << len << endl;

    ofstream fout("string.txt");
    fout << str;
    fout.close();

    ifstream fin("string.txt");
    char ch;
    cout << "File content: ";

    while(fin.get(ch)) {
      cout << ch;
    }

    fin.close();
    return 0;
}
```

Start here | ques2.cpp | ques1.cpp | NOTES.TXT | ques1actual.cpp | NUM.TXT | ques3.cpp | ques5.cpp | source.txt | dest.txt | ues4.cpp | string.txt

```
1   Hello Kashish Aggarwal
```

```
"C:\Users\priya\OneDrive\Des
Enter a string: Hello Kashish Aggarwal
Length = 22
File content: Hello Kashish Aggarwal
Process returned 0 (0x0)   execution time : 8.372 s
Press any key to continue.
```

5. Write a program to do the followings:

(a) Create a file A-Z and read 10th character using seekg().

(b) Overwrite 5th number using seekp().

(c) Find file size using tellg().

(d) Read last character using seekg().

(e) Create a file data.txt with multiple lines. Read the file and:

i. Move the pointer to the 10th byte using seekg()

ii. Display the current position using tellg().

iii. Read and display the remaining content.

Sol 5:

```
#include <iostream>

#include <fstream>

using namespace std;


int main() {


  // (a) Create file with A-Z and read 10th character

  ofstream fout("file.txt");

  for(char ch = 'A'; ch <= 'Z'; ch++) {

    fout << ch;

  }
```

```
fout.close();

ifstream fin("file.txt");
fin.seekg(9, ios::beg);   // 10th character (index starts from 0)
char ch;
fin.get(ch);
cout << "(a) 10th character: " << ch << endl;
fin.close();

// (b) Overwrite 5th character using seekp()
fstream file("file.txt", ios::in | ios::out);
file.seekp(4, ios::beg);  // 5th position
file.put('X');        // overwrite with 'X'
file.close();

// Display updated content
fin.open("file.txt");
cout << "(b) Updated content: ";
while(fin.get(ch)) cout << ch;
cout << endl;
fin.close();

// (c) Find file size using tellg()
fin.open("file.txt");
fin.seekg(0, ios::end);   // go to end
cout << "(c) File size: " << fin.tellg() << " bytes" << endl;
```

```
fin.close();

// (d) Read last character using seekg()
fin.open("file.txt");
fin.seekg(-1, ios::end);  // last character
fin.get(ch);
cout << "(d) Last character: " << ch << endl;
fin.close();

// (e) Create data.txt with multiple lines
ofstream data("data.txt");
data << "Hello World\n";
data << "This is a test file\n";
data << "C++ File Handling Example\n";
data.close();

ifstream dfile("data.txt");

// (i) Move pointer to 10th byte
dfile.seekg(9, ios::beg);

// (ii) Display current position
cout << "(e-ii) Current position: " << dfile.tellg() << endl;

// (iii) Read remaining content
cout << "(e-iii) Remaining content:\n";
```

```
  while(dfile.get(ch)) {
    cout << ch;
  }

  dfile.close();

  return 0;
}
```

Start here | ques5.cpp | string.txt | file.txt | data.txt

```
ABCDXFGHIJKLMNOPQRSTUVWXYZ
```

Start here | ques5.cpp | string.txt | file.txt | data.txt

```
Hello World
This is a test file
C++ File Handling Example
```

"C:\Users\priya\OneDrive\Des

```
(a) 10th character: J
(b) Updated content: ABCDXFGHIJKLMNOPQRSTUVWXYZ
(c) File size: 26 bytes
(d) Last character: Z
(e-ii) Current position: 9
(e-iii) Remaining content:
ld
This is a test file
C++ File Handling Example

Process returned 0 (0x0)   execution time : 0.243 s
Press any key to continue.
```

6. Write a program to do the followings:

i. Create a file with text "Hello World"

ii. After writing each character, display the current position of the put pointer using tellp().

iii. Replace "World" with "C++" using random access

Sol6:

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
  fstream file("hello.txt", ios::out);

  string text = "Hello World";

  for(int i = 0; i < text.length(); i++) {
    file.put(text[i]);
    cout << "Position: " << file.tellp() << endl;
  }

  file.close();

  // Replace "World" with "C++"
  fstream file2("hello.txt", ios::in | ios::out);
  file2.seekp(6, ios::beg); // start of "World"
  file2 << "C++  ";
```

```
    file2.close();

    cout << "Replacement done!";
    return 0;
}
```

Start here | ques5.cpp | string.txt | file.txt | data.txt | ques6.cpp | hello.txt

```
Hello C++
```

"C:\Users\priya\OneDrive\Des

```
Position: 1
Position: 2
Position: 3
Position: 4
Position: 5
Position: 6
Position: 7
Position: 8
Position: 9
Position: 10
Position: 11
Replacement done!
Process returned 0 (0x0)   execution time : 0.164 s
Press any key to continue.
```